Fireside

MIXTURE OF EXPERTS (MoE)

Deepseek V3

Lakshit Talreja

Why MoE?

Feature	DeepSeek-V3 / R1	ChatGPT (GPT-4 / 4o)
Estimated Training Cost	**~$5.58 Million**	**$100 Million+**
Total Parameters	671 Billion (MoE)	Reported ~1.8 Trillion
Active Parameters	37 Billion (per token)	Not disclosed (Estimated 280B+)
Training Duration	**~55 Days**	Estimated 3–6 Months
Hardware Used	~2,000 NVIDIA H800 GPUs	Estimated 10,000+ H100 GPUs

deepseek

The first appearance of the Mixture of Experts (MoE) architecture was in 1991 with the seminal paper titled "Adaptive Mixtures of Local Experts".

What is MoE?

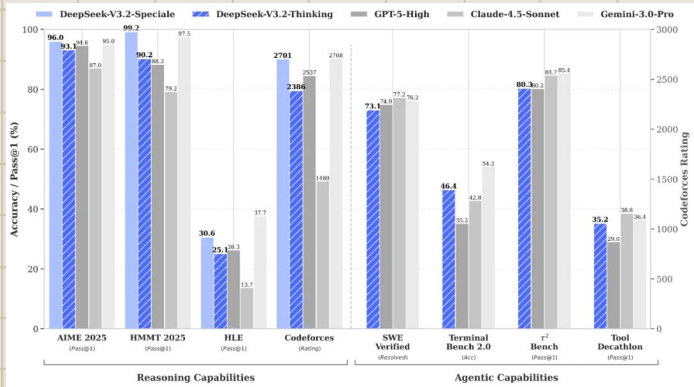
MoE is a technique where specialised sub networks(experts) are activated instead of activating the whole model.

What are Experts

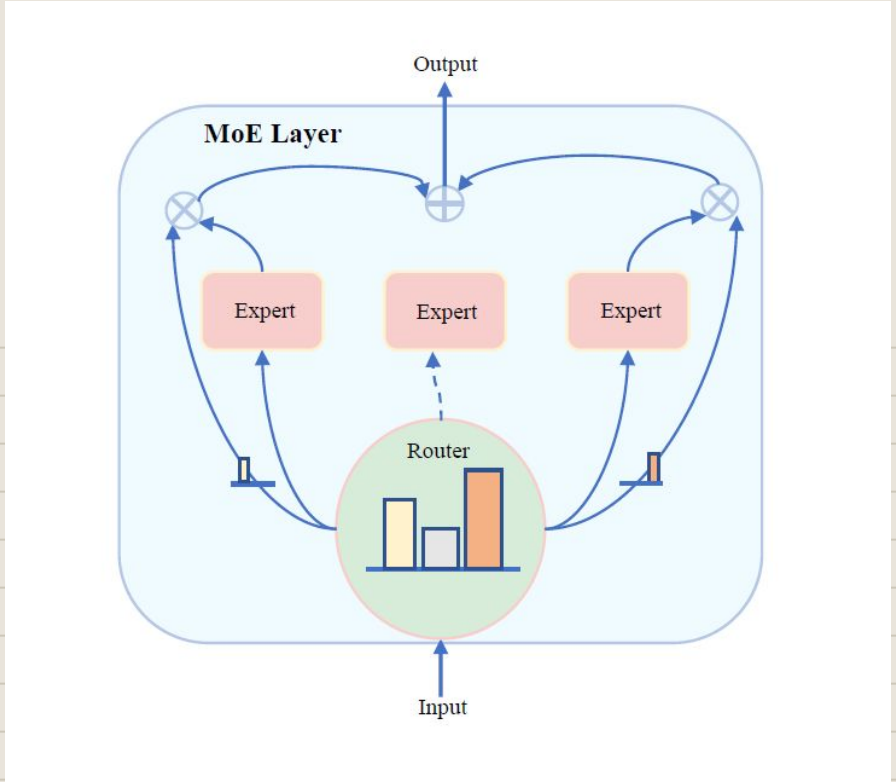
In a tradtional AI model, the model acts as a single complete brain whereas in MoE the brain is essetially spit into sub parts each tasked with handling only one task. Each of these subparts can be considered to be experts.

So in a MoE model, the experts are each tasked with solving or answering to certain inputs and are only activated when they are needed unlike the traditional model where the whole model is activated when it is given some input.

What did Deepseek do differently



Basic Design of MoE



Router / Gating Function

- The function should be able to accurately discern the characteristics of both the input and expert data. This allows it to assign the right data to the right expert, ensuring that each expert recieves a suffciently large and cohert dataset.

- The data should be evenly distributed to avoid model collapse.

Linear Gating

$$G(x)_i = \text{softmax}(\text{TopK}(g(x) + \mathcal{R}_{\text{noise}}, k))_i, \quad (1)$$

$$\text{TopK}(v) = \begin{cases} v_i & \text{if } v_i \text{ is one of the top } k \text{ elements,} \\ -\infty & \text{otherwise.} \end{cases} \quad (2)$$

Non linear Gating

$$G(x) = \text{TopK}\left(\text{softmax}\left(\frac{E^T W_{linear} x}{\tau \|W_{linear} x\| \|E\|}\right)\right) \quad (3)$$

$$\frac{A \cdot B}{\|A\| \|B\|}$$

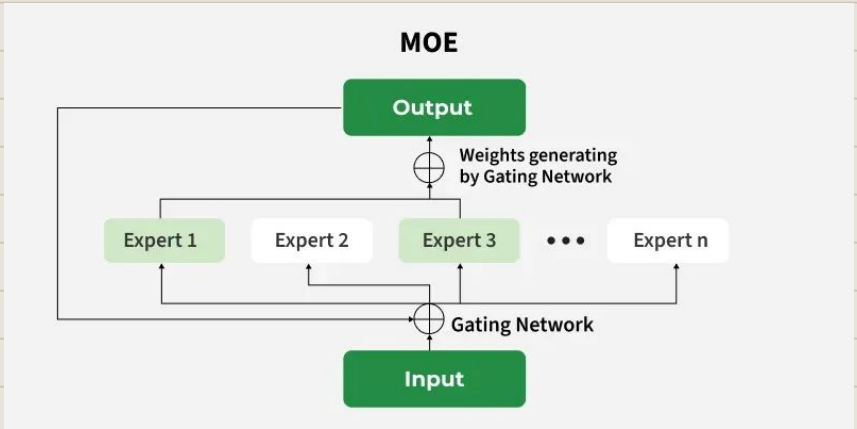
Sigmoid Gating

$$G(x)_i = \sigma(W_i x)$$

Expert Networks

Replace the FFN layer in Transformer with an MoE layer.

Apply MoE to the attention module in Transformer



Routing strategy

Most classical routing strategy, it uses token representations to determine the routing decisions. We mainly deal with two types of tokens, text tokens and patch (image) tokens.

Text tokens

MaskMoE and RNN is used.
Experts are hidden from the words.

Patch tokens

Image is split into patches and routes each patch and sorts in descending order

Model Collapse

1. **Rich get richer**

   Few experts are regularly activated and end up training well but the ones that are never activated lead to an imbalance in expert utilization causing a collapse

2. **Expert Capacity**

   It is the limit of data that a expert can handle, basically we don't want to overwhelm our experts with too much data.
   If the Expert is **full** then the tokens are dropped out or passed without being processed.

3. **Switch Transformer**

$$\text{loss} = \alpha \cdot N \cdot \sum_{i=1}^{N} f_i \cdot Q_i.$$

NEXT STEPS

Reading Hierarchical MoE

https://lakshittalreja.github.io/posts/moe/

THANK YOU!